

AIML ENGINEERING LAB | P053

DRAM Memory Yield Predictor 40-Day Production Learning Report

Interview-ready and PDF-ready operating record for a full MLOps pipeline: 16M-row training, HybridTransformerCNN, Airflow, Kafka, Spark, MLflow, S3, AWS GPU retraining, canary evaluation, rollback, incident recovery, and cost discipline.

What this document is

A living HTML report that can be regenerated as future production days complete. It starts with high-level KPIs, then gives the full day-by-day ledger, followed by narrative learnings, issues, fixes, decisions, cost, and interview talking points.

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1. Executive KPIs

The project demonstrates an end-to-end production MLOps system for DRAM yield prediction. The key point for interviews: this is not just a model result; it is a full operational lifecycle with drift detection, retraining, canary promotion, rollback, cloud cost control, and failure recovery.

PRODUCTION SIMULATION

40 days

Feb 20 - Mar 31, 2026

ROWS PROCESSED

200,000,000

5,000,000 rows/day

PARQUET OUTPUT

8.34 GB

8,542.2 MB total

WALL CLOCK

219.4 min

End-to-end 40-day A100 simulation

DRIFT REPORTS

32

24 critical drift days

RETRAINS

1

Day 30 retrain -> v2

ROLLBACKS

1

Day 39 canary failure rollback

S3 UPLOADS

72

Daily artifacts persisted

CHAMPION PARAMS

317,633

A100 val AUC-PR 0.0536

INFRA RECOVERY

85 GB free

EC2 disk after Docker volume cleanup

Headline: 200M rows were processed across a 40-day production simulation. Drift accumulated from Day 17, the staleness gate opened on Day 30, v2 was retrained and promoted, and a Day 39 canary failure correctly rolled back to v2.

2. System and Model

Dataset and Signal

Training rows	10,000,000
Columns	37 total; 29 numeric; 7 categorical
Fail rate	0.617%
Label noise	12,464 noisy labels
Missing cells	3,796,537 (1.03%)
Spatial effect	Edge fail rate 1.154% vs center 0.353%; ratio 3.3x

Training and Model

Architecture	HybridTransformerCNN
Parameters	317,633
A100 GPU	NVIDIA A100-SXM4-40GB, 39.5 GB VRAM
A100 throughput	88,128 samples/sec
T4 throughput	18,868 samples/sec
Mixed precision	bfloat16 on A100; float16 + GradScaler on T4 when stable

Model Metrics

Split	AUC-PR	AUC-ROC	F1	Recall	Precision	TP	FP	FN	TN
A100 Val	0.0536	0.8157	0.1272	21.6%	9.0%	3,374	34,075	12,221	1,950,330
A100 Test	0.0492	0.7994	0.1207	17.3%	9.3%	2,278	22,328	10,873	1,964,521
A100 Unseen	0.0565	0.8148	0.1308	23.1%	9.1%	3,868	38,541	12,853	1,944,738
T4 Val	0.0524	0.8100	0.1249	21.1%	8.9%	3,294	33,870	12,301	1,950,535

Generate

Daily DRAM production rows

Stream

Kafka producer/consumer path

Transform

Spark ETL to Parquet/S3

Monitor

PSI drift and staleness gate

React

GPU retrain, canary, promote/rollback

3. 40-Day Master Table

This table is the operational ledger: rows, data volume, drift counts, model version, gate outcome, artifacts, and runtime for every day.

Day	Date	Rows	Parquet MB	Scenario	Model	Crit	Warn	Max PSI	Dominant PSI Feature	Gate
1	2026-02-20	5,000,000	214.1	steady	v1_original	-	-	-	-	Baseline
2	2026-02-21	5,000,000	213.7	steady	v1_original	-	-	-	-	Baseline
3	2026-02-22	5,000,000	213.7	steady	v1_original	-	-	-	-	Baseline
4	2026-02-23	5,000,000	214.1	steady	v1_original	-	-	-	-	Baseline
5	2026-02-24	5,000,000	214.1	steady	v1_original	-	-	-	-	Baseline
6	2026-02-25	5,000,000	213.7	steady	v1_original	-	-	-	-	Baseline
7	2026-02-26	5,000,000	214.0	steady	v1_original	-	-	-	-	Baseline
8	2026-02-27	5,000,000	214.0	steady	v1_original	-	-	-	-	Baseline
9	2026-02-28	5,000,000	213.4	false alarm	v1_original	2	0	2.05	test_temp_c (2.05)	Clear
10	2026-03-01	5,000,000	214.0	auto recover	v1_original	0	0	0.08	test_temp_c (0.08)	Clear
11	2026-03-02	5,000,000	214.0	gradual drift	v1_original	0	0	0.05	test_temp_c (0.05)	Clear
12	2026-03-03	5,000,000	213.9	gradual drift	v1_original	0	0	0.06	gate_oxide_thickness_a (0.06)	Clear
13	2026-03-04	5,000,000	214.1	gradual drift	v1_original	0	1	0.14	gate_oxide_thickness_a (0.14)	Warning
14	2026-03-05	5,000,000	214.1	gradual drift	v1_original	1	0	0.24	gate_oxide_thickness_a (0.24)	Clear
15	2026-03-06	5,000,000	214.1	gradual drift	v1_original	1	0	0.37	gate_oxide_thickness_a (0.37)	Clear
16	2026-03-07	5,000,000	214.1	gradual drift	v1_original	1	0	0.53	gate_oxide_thickness_a (0.53)	Clear
17	2026-03-08	5,000,000	213.5	gradual drift	v1_original	3	0	0.72	gate_oxide_thickness_a (0.72)	Baseline
18	2026-03-09	5,000,000	213.7	gradual drift	v1_original	3	0	0.92	gate_oxide_thickness_a (0.92)	Baseline
19	2026-03-10	5,000,000	213.5	sudden shift	v1_original	3	1	1.45	retention_time_ms (1.45)	Baseline
20	2026-03-11	5,000,000	213.5	threshold 1	v1_original	3	2	1.49	retention_time_ms (1.49)	Baseline

Day	Date	Rows	Parquet MB	Scenario	Model	Crit	Warn	Max PSI	Dominant PSI Feature	Gate
21	2026-03-12	5,000,000	213.5	continued drift	v1_original	3	1	1.71	gate_oxide_thickness_a (1.71)	BI
22	2026-03-13	5,000,000	213.2	continued drift	v1_original	4	0	2.30	retention_time_ms (2.30)	BI
23	2026-03-14	5,000,000	213.3	continued drift	v1_original	4	0	2.42	retention_time_ms (2.42)	BI
24	2026-03-15	5,000,000	213.2	continued drift	v1_original	4	1	2.99	retention_time_ms (2.99)	BI
25	2026-03-16	5,000,000	213.2	continued drift	v1_original	4	1	2.97	gate_oxide_thickness_a (2.97)	BI
26	2026-03-17	5,000,000	213.0	threshold 2	v1_original	5	0	4.48	retention_time_ms (4.48)	BI
27	2026-03-18	5,000,000	213.3	worsening	v1_original	5	0	3.68	gate_oxide_thickness_a (3.68)	BI
28	2026-03-19	5,000,000	213.0	worsening	v1_original	5	0	4.41	retention_time_ms (4.41)	BI
29	2026-03-20	5,000,000	213.1	worsening	v1_original	5	0	4.90	retention_time_ms (4.90)	BI
30	2026-03-21	5,000,000	213.0	worsening	v2_retrained_day30	5	0	5.25	retention_time_ms (5.25)	Re
31	2026-03-22	5,000,000	213.0	retrain trigger	v2_retrained_day30	5	0	5.81	retention_time_ms (5.81)	BI
32	2026-03-23	5,000,000	213.0	post retrain recovery	v2_retrained_day30	4	0	5.25	gate_oxide_thickness_a (5.25)	BI
33	2026-03-24	5,000,000	213.0	post retrain recovery	v2_retrained_day30	4	0	6.03	retention_time_ms (6.03)	BI
34	2026-03-25	5,000,000	213.0	post retrain recovery	v2_retrained_day30	4	0	5.77	retention_time_ms (5.77)	BI
35	2026-03-26	5,000,000	213.0	post retrain recovery	v2_retrained_day30	4	0	5.18	gate_oxide_thickness_a (5.18)	BI
36	2026-03-27	5,000,000	213.6	second drift	v2_retrained_day30	4	0	6.32	retention_time_ms (6.32)	BI
37	2026-03-28	5,000,000	213.6	second drift	v2_retrained_day30	4	1	6.09	retention_time_ms (6.09)	BI
38	2026-03-29	5,000,000	213.6	second drift	v2_retrained_day30	5	0	5.81	gate_oxide_thickness_a (5.81)	BI

Day	Date	Rows	Parquet MB	Scenario	Model	Crit	Warn	Max PSI	Dominant PSI Feature	Gate
39	2026-03-30	5,000,000	213.6	bad model deploy	v2_retrained_day30	5	0	6.05	retention_time_ms (6.05)	Red
40	2026-03-31	5,000,000	213.7	final recovery	v2_retrained_day30	5	0	5.92	gate_oxide_thickness_a (5.92)	Blue

Scenario Mix

Scenario	Days
steady	8
gradual drift	8
continued drift	5
worsening	4
post retrain recovery	4
second drift	3
false alarm	1
auto recover	1
sudden shift	1
threshold 1	1
threshold 2	1
retrain trigger	1
bad model deploy	1
final recovery	1

Event Counts

Event	Count
s3_uploaded_2_files	32
drift_critical_4_features	10
drift_critical_5_features	9
s3_uploaded_1_files	8
drift_clean	7
drift_critical_3_features	5
drift_warning_1_features	1
retrain_blocked_staleness_17d	1
retrain_blocked_staleness_18d	1
retrain_blocked_staleness_19d	1
retrain_blocked_staleness_20d	1
retrain_blocked_staleness_21d	1
retrain_blocked_staleness_22d	1
retrain_blocked_staleness_23d	1

4. Day-by-Day Story

The audit table gives the numbers; these notes explain what happened, why each stage mattered, and what a reviewer should understand from the progression.

Day 1 2026-02-20

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 2 2026-02-21

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 213.7 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 3 2026-02-22

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 213.7 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 4 2026-02-23

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 5 2026-02-24

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 6 2026-02-25

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 213.7 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 7 2026-02-26

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 214.0 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 8 2026-02-27

Baseline

Normal production baseline

Data was generated and uploaded with no drift report required in the warm-up window.

Numbers: Rows: 5,000,000 | Parquet: 214.0 MB | Model: v1_original | S3 files: 1

Events: s3_uploaded_1_files

Day 9 2026-02-28

Clean

False-alarm drift test

Two features spiked, but the retrain gate held because the overall condition was not strong enough for promotion risk.

Numbers: Rows: 5,000,000 | Parquet: 213.4 MB | Model: v1_original | S3 files: 2 | Critical features: 2 | Warning features: 0 | Max PSI: 2.05

Events: drift_clean, s3_uploaded_2_files

Day 10 2026-03-01

Clean

Auto-recovery check

The previous abnormal shift disappeared, confirming the monitor can distinguish transient movement from durable drift.

Numbers: Rows: 5,000,000 | Parquet: 214.0 MB | Model: v1_original | S3 files: 2 | Critical features: 0 | Warning features: 0 | Max PSI: 0.08

Events: drift_clean, s3_uploaded_2_files

Day 11 2026-03-02

Clean

Gradual drift accumulation

Drift increased slowly; 0 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 214.0 MB | Model: v1_original | S3 files: 2 | Critical features: 0 | Warning features: 0 | Max PSI: 0.05

Events: drift_clean, s3_uploaded_2_files

Day 12 2026-03-03

Clean

Gradual drift accumulation

Drift increased slowly; 0 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 213.9 MB | Model: v1_original | S3 files: 2 | Critical features: 0 | Warning features: 0 | Max PSI: 0.06

Events: drift_clean, s3_uploaded_2_files

Day 13 2026-03-04

Warning

Gradual drift accumulation

Drift increased slowly; 0 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 2 | Critical features: 0 | Warning features: 1 | Max PSI: 0.14

Events: drift_warning_1_features, s3_uploaded_2_files

Day 14 2026-03-05

Clean

Gradual drift accumulation

Drift increased slowly; 1 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 2 | Critical features: 1 | Warning features: 0 | Max PSI: 0.24

Events: drift_clean, s3_uploaded_2_files

Day 15 2026-03-06

Clean

Gradual drift accumulation

Drift increased slowly; 1 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 2 | Critical features: 1 | Warning features: 0 | Max PSI: 0.37

Events: drift_clean, s3_uploaded_2_files

Day 16 2026-03-07

Clean

Gradual drift accumulation

Drift increased slowly; 1 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 214.1 MB | Model: v1_original | S3 files: 2 | Critical features: 1 | Warning features: 0 | Max PSI: 0.53

Events: drift_clean, s3_uploaded_2_files

Day 17 2026-03-08

Blocked by staleness

Gradual drift accumulation

Drift increased slowly; 3 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 213.5 MB | Model: v1_original | S3 files: 2 | Critical features: 3 | Warning features: 0 | Max PSI: 0.72

Events: drift_critical_3_features, retrain_blocked_staleness_17d, s3_uploaded_2_files

Day 18 2026-03-09

Blocked by staleness

Gradual drift accumulation

Drift increased slowly; 3 critical features were tracked without immediate retraining while the staleness gate matured.

Numbers: Rows: 5,000,000 | Parquet: 213.7 MB | Model: v1_original | S3 files: 2 | Critical features: 3 | Warning features: 0 | Max PSI: 0.92

Events: drift_critical_3_features, retrain_blocked_staleness_18d, s3_uploaded_2_files

Day 19 2026-03-10

Blocked by staleness

Persistent process drift

The pipeline recorded 3 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.5 MB | Model: v1_original | S3 files: 2 | Critical features: 3 | Warning features: 1 | Max PSI: 1.45

Events: drift_critical_3_features, retrain_blocked_staleness_19d, s3_uploaded_2_files

Day 20 2026-03-11

Blocked by staleness

Persistent process drift

The pipeline recorded 3 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.5 MB | Model: v1_original | S3 files: 2 | Critical features: 3 | Warning features: 2 | Max PSI: 1.49

Events: drift_critical_3_features, retrain_blocked_staleness_20d, s3_uploaded_2_files

Day 21 2026-03-12

Blocked by staleness

Persistent process drift

The pipeline recorded 3 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.5 MB | Model: v1_original | S3 files: 2 | Critical features: 3 | Warning features: 1 | Max PSI: 1.71

Events: drift_critical_3_features, retrain_blocked_staleness_21d, s3_uploaded_2_files

Day 22 2026-03-13

Blocked by staleness

Persistent process drift

The pipeline recorded 4 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.2 MB | Model: v1_original | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 2.30

Events: drift_critical_4_features, retrain_blocked_staleness_22d, s3_uploaded_2_files

Day 23 2026-03-14

Blocked by staleness

Persistent process drift

The pipeline recorded 4 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.3 MB | Model: v1_original | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 2.42

Events: drift_critical_4_features, retrain_blocked_staleness_23d, s3_uploaded_2_files

Day 24 2026-03-15

Blocked by staleness

Persistent process drift

The pipeline recorded 4 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.2 MB | Model: v1_original | S3 files: 2 | Critical features: 4 | Warning features: 1 | Max PSI: 2.99

Events: drift_critical_4_features, retrain_blocked_staleness_24d, s3_uploaded_2_files

Day 25 2026-03-16

Blocked by staleness

Persistent process drift

The pipeline recorded 4 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.2 MB | Model: v1_original | S3 files: 2 | Critical features: 4 | Warning features: 1 | Max PSI: 2.97

Events: drift_critical_4_features, retrain_blocked_staleness_25d, s3_uploaded_2_files

Day 26 2026-03-17

Blocked by staleness

Persistent process drift

The pipeline recorded 5 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v1_original | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 4.48

Events: drift_critical_5_features, retrain_blocked_staleness_26d, s3_uploaded_2_files

Day 27 2026-03-18

Blocked by staleness

Persistent process drift

The pipeline recorded 5 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.3 MB | Model: v1_original | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 3.68

Events: drift_critical_5_features, retrain_blocked_staleness_27d, s3_uploaded_2_files

Day 28 2026-03-19

Blocked by staleness

Persistent process drift

The pipeline recorded 5 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v1_original | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 4.41

Events: drift_critical_5_features, retrain_blocked_staleness_28d, s3_uploaded_2_files

Day 29 2026-03-20

Blocked by staleness

Persistent process drift

The pipeline recorded 5 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.1 MB | Model: v1_original | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 4.90

Events: drift_critical_5_features, retrain_blocked_staleness_29d, s3_uploaded_2_files

Day 30 2026-03-21

Retrain

Persistent process drift

The pipeline recorded 5 critical features and preserved the evidence needed for the Day 30 retrain decision.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 5.25

Events: drift_critical_5_features, RETRAIN_TRIGGERED, s3_uploaded_2_files

Day 31 2026-03-22

Blocked by staleness

Post-promotion guardrail

The newly promoted v2 model served inference; retraining was blocked again because the model age was only one day.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 5.81

Events: drift_critical_5_features, retrain_blocked_staleness_1d, s3_uploaded_2_files

Day 32 2026-03-23

Blocked by staleness

Recovery monitoring

The system monitored whether v2 reduced production risk while continuing to tag residual drift.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 5.25

Events: drift_critical_4_features, retrain_blocked_staleness_2d, s3_uploaded_2_files

Day 33 2026-03-24

Blocked by staleness

Recovery monitoring

The system monitored whether v2 reduced production risk while continuing to tag residual drift.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 6.03

Events: drift_critical_4_features, retrain_blocked_staleness_3d, s3_uploaded_2_files

Day 34 2026-03-25

Blocked by staleness

Recovery monitoring

The system monitored whether v2 reduced production risk while continuing to tag residual drift.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 5.77

Events: drift_critical_4_features, retrain_blocked_staleness_4d, s3_uploaded_2_files

Day 35 2026-03-26

Blocked by staleness

Recovery monitoring

The system monitored whether v2 reduced production risk while continuing to tag residual drift.

Numbers: Rows: 5,000,000 | Parquet: 213.0 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 5.18

Events: drift_critical_4_features, retrain_blocked_staleness_5d, s3_uploaded_2_files

Day 36 2026-03-27

Blocked by staleness

Second drift wave

A second wave of drift was observed, but the staleness gate prevented excessive retraining.

Numbers: Rows: 5,000,000 | Parquet: 213.6 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 0 | Max PSI: 6.32

Events: drift_critical_4_features, retrain_blocked_staleness_6d, s3_uploaded_2_files

Day 37 2026-03-28

Blocked by staleness

Second drift wave

A second wave of drift was observed, but the staleness gate prevented excessive retraining.

Numbers: Rows: 5,000,000 | Parquet: 213.6 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 4 | Warning features: 1 | Max PSI: 6.09

Events: drift_critical_4_features, retrain_blocked_staleness_7d, s3_uploaded_2_files

Day 38 2026-03-29

Blocked by staleness

Second drift wave

A second wave of drift was observed, but the staleness gate prevented excessive retraining.

Numbers: Rows: 5,000,000 | Parquet: 213.6 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 5.81

Events: drift_critical_5_features, retrain_blocked_staleness_8d, s3_uploaded_2_files

Day 39 2026-03-30

Rollback

Canary rollback test

A deliberately bad candidate failed canary evaluation and the system rolled back to v2 automatically.

Numbers: Rows: 5,000,000 | Parquet: 213.6 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 6.05

Events: drift_critical_5_features, retrain_blocked_staleness_9d, BAD_MODEL_DEPLOYED, CANARY_FAILED, ROLLBACK_TO_v2, s3_uploaded_2_files

Day 40 2026-03-31

Blocked by staleness

Recovered final state

The production run finished with v2 active and rollback behavior proven.

Numbers: Rows: 5,000,000 | Parquet: 213.7 MB | Model: v2_retrained_day30 | S3 files: 2 | Critical features: 5 | Warning features: 0 | Max PSI: 5.92

Events: drift_critical_5_features, retrain_blocked_staleness_10d, SYSTEM_RECOVERED, s3_uploaded_2_files

5. Live AWS Recovery and Current Production Learning

Why this section exists: The clean 40-day simulation proves the design, while the July AWS daily run shows production reality: timeout budgets, Docker disk growth, Spark memory pressure, S3 state, retry hygiene, schedule guard mistakes, and final cloud shutdown. These are strong interview stories because they show operating judgment, not only modeling skill.

Date	Area	Symptom	Root Cause	Fix	Prevention
2026-07-01	Day 30 retrain orchestration	Day 30 drift was real, but the retrain DAG did not complete and the champion stayed on Day 1.	TriggerDagRunOperator triggered the retrain DAG without waiting for completion; EC2 cleanup could stop the host while retraining was still expected.	Added <code>wait_for_completion=True</code> and <code>poke_interval=60</code> ; retrain results now upload to S3 and update <code>pipeline_state.json</code> on canary pass.	Production state is updated only after canary-confirmed promotion; retrain and canary artifacts are stored in S3.
2026-07-01	GitHub Actions timeout	Manual Day 30 re-run was cancelled around the 260-minute mark.	Workflow timeout was too short for a full GPU retrain day.	Raised <code>daily_pipeline.yml</code> <code>timeout-minutes</code> to 480.	Retrain days now have an 8-hour envelope for ETL, training, canary, artifact sync, and cleanup.
2026-07-03	Spark ETL memory pressure	Day 31 failed with Airflow zombie detection for <code>spark_etl</code> .	Spark local[2] with a 2 GB driver plus Docker services exceeded practical memory headroom on <code>g4dn.xlarge</code> .	Changed Spark tasks to local[1], 1 GB driver memory, <code>maxResultSize=256m</code> , <code>shuffle.partitions=4</code> , <code>retries=2</code> , and <code>execution_timeout=45m</code> .	Spark now uses a smaller footprint and Airflow has retry/timeout controls instead of waiting on dead processes.
2026-07-04	EC2 disk saturation	Day 31 cron failed during source sync with tar: No space left on device.	Anonymous Docker volumes accumulated Kafka/Airflow data across repeated daily runs and consumed 78 GB; root volume reached 100%.	Deleted 26 anonymous Docker volumes, cleared local data already uploaded to S3, restarted <code>containerd/Docker</code> , and pruned Docker resources.	Created <code>/etc/cron.daily/docker-cleanup</code> and kept S3 as the durable artifact store.
2026-07-04	Day 32 accidental retrain trigger	Day 32 finished ETL and drift detection, then blocked in <code>trigger_retrain</code> while the child retrain DAG stayed queued.	The daily gate only checked <code>day >= 30</code> , so drift after Day 30 could retrigger training immediately; the retrain DAG was also paused, leaving <code>TriggerDagRunOperator</code> waiting.	Marked the accidental Day 32 child retrain run successful to unblock the parent daily DAG, then added a <code>champion_updated_day</code> cooldown read from S3 <code>pipeline_state.json</code> .	Days after a champion promotion now skip retrain until the model reaches the configured staleness window; queued child retrains are no longer part of normal Day 31-39 flow.
2026-07-04	Kafka container memory	Kafka exited with code 137 during the Day 32 recovery window and Day 33 memory climbed toward the 1 GB container cap.	The Confluent Kafka JVM had a container memory limit but no explicit heap cap, so the broker could be killed under publish pressure.	Raised the live Day 33 Kafka container limit to 1.5 GB and committed <code>KAFKA_HEAP_OPTS=-Xms256m -Xmx512m</code> with restart: unless-stopped for future runs.	Future dispatches run Kafka with bounded heap and automatic restart behavior while leaving enough host memory for Airflow and Spark.
2026-07-11	Final Day 40 closure	The live AWS daily pipeline completed all intended 40 days, but project reports and trackers still reflected the Day 33 interim recovery point.	The recovery report was intentionally generated mid-run and had not yet been refreshed after the scheduled Day 34-40 GitHub Actions runs completed.	Audited S3 <code>pipeline_state.json</code> , Day 29-40 artifacts, GitHub Actions run history, and AWS resource state; refreshed the report to Day 40 complete with the Day 30 v2 champion still active.	Treat S3 <code>pipeline_state.json</code> as the final source of truth before publishing status documents or cost-closeout notes.
2026-07-17	Post-Day40 schedule guard	Scheduled GitHub Actions runs continued after Day 40 and generated real Day 41-45 Parquet, drift,	The completion check used <code>exit 0</code> inside one step, but GitHub Actions continued to later steps that started RDS/EC2 and triggered Airflow.	Disabled the cron schedule and added a <code>should_run</code> guard to every AWS-starting or Airflow-triggering workflow step.	Future runs require explicit <code>workflow_dispatch</code> , and <code>complete/day>40</code> state skips expensive AWS steps instead of only logging completion.

Date	Area	Symptom	Root Cause	Fix	Prevention
		and summary artifacts.			

Current State Snapshot

Intended production scope	Day 1-40 complete; Day 30 v2 champion retained through the intended run.
Current S3 pipeline state	complete; current_day=46; last_completed_day=45; last_run=2026-07-16T02:05:57Z
Post-Day40 finding	Days 41-45 were real unintended scheduled runs caused by a workflow guard bug; cron is now disabled and AWS-starting steps are guarded by should_run.
Champion model in pipeline_state.json	s3://p053-mlflow-artifacts/models/day30_v2_retrained.pt
Artifact status	Days 29-45 have current-prefix production Parquet, drift reports, and summaries on S3; Days 41-45 are extra artifacts from the schedule leak.
Known live issues fixed	Spark zombie/OOM settings reduced; EC2 disk recovered from 100% full; missing reference data now restores from S3; Day 32 accidental retrain waits are blocked by champion cooldown; Kafka heap is capped and restartable; post-Day40 workflow schedule is disabled.
AWS shutdown status	EC2 g4dn.xlarge stopped; RDS db.t3.micro stopped; no NAT gateways available or pending.
Residual billable resources	125 GiB gp3 EBS volume; 20 GiB RDS storage; 21 automated RDS snapshots; ~4.80 GB current S3 objects with 387 versions; 1 ECR repo; 1 associated public IPv4/EIP.
Retention decision	Keep minimal retained AWS evidence resources for now. Day 41-45 extra artifacts can be deleted separately if desired.

6. Cost and Value Ledger

The cost story is deliberately practical: use A100 only when its throughput is justified, use T4 for the 317K-parameter production retrains, stop idle RDS/EC2, and keep S3 as durable storage.

Cost Item	Unit Cost / Estimate	Role	Planning Note
AWS EC2 g4dn.xlarge	\$0.526/hour	T4 GPU host for Airflow-triggered retrains and production runs	~\$2.10 for 4 hours; ~\$52.60 for 100 hours
AWS RDS db.t3.micro	~\$0.018/hour	PostgreSQL backend for MLflow when AWS is active	~\$13/month if left running; stopped between daily runs to reduce idle cost
S3 artifacts	Low single-digit USD/month at this scale	Models, Parquet, drift reports, metrics, checkpoints	40-day Parquet footprint in simulation: ~8.5 GB plus models/JSON
Final residual AWS resources	Billable until deleted or expired	125 GiB gp3 EBS; 20 GiB RDS; 21 automated RDS snapshots; ~4.80 GB current S3 data with 387 object versions; 1 ECR repo; 1 associated public IPv4/EIP	EC2/RDS/NAT compute is stopped. User chose to keep minimal evidence resources for now; Day 41-45 cleanup remains optional.
Colab fallback T4	~1.36 CU/hour	Fallback retraining if AWS GPU is unavailable	100 CU pack was the budget choice for multiple T4 runs
Colab A100	~6.79 CU/hour	Initial high-speed training and large simulation fallback	Useful for >1 TB/day or heavy experimentation
Estimated business upside	\$36M/year	From avoided escaped DRAM defects	67 defects/month avoided x \$45K x 12 months

7. Engineering Decisions and Interview Talking Points

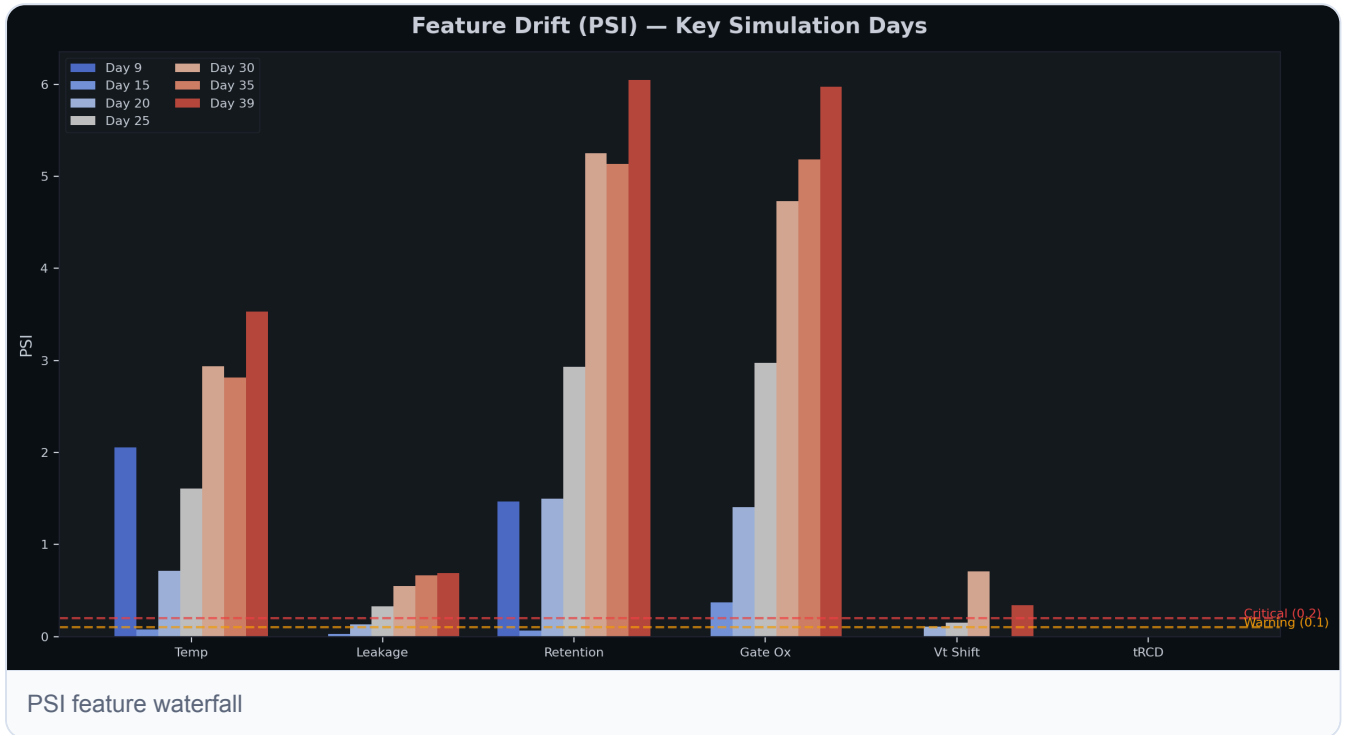
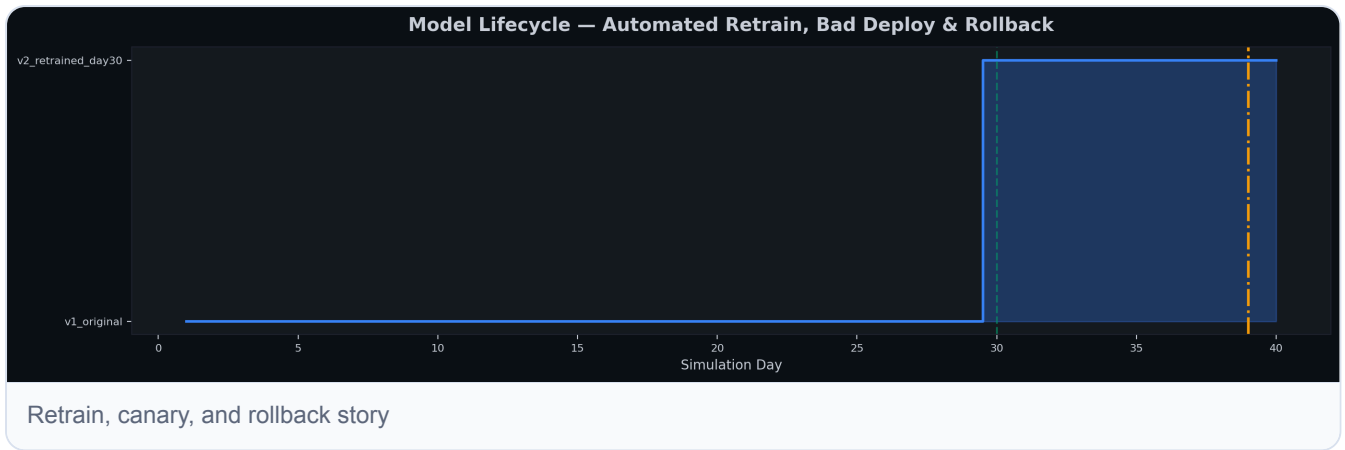
Decision	Why It Matters
AUC-PR as primary metric	At roughly 1:159 defect imbalance, AUC-ROC can look healthy while missing the minority class. AUC-PR directly measures ranking quality for defects.
HybridTransformerCNN	Per-feature tokenization models heterogeneous DRAM parameters; CNN layers capture spatial wafer patterns; fusion MLP combines process, test, and location signals.
Focal loss	The model must learn from rare defects without synthetic oversampling noise. Focal loss emphasizes hard minority examples.
bfloat16 on A100	float16 with GradScaler collapsed under focal-loss gradients; bfloat16 preserves exponent range and trained stably without GradScaler.
T4 for production retrains	The 317K-parameter model fits comfortably on T4. A100 is reserved for very large training/data volumes.
Staleness gate	Early drift is tagged and monitored, but retraining is blocked until the model has enough production age to avoid thrashing.
Canary before promotion	A retrained model becomes champion only after canary evaluation; Day 39 deliberately tested rollback behavior.
S3 as durable system of record	Local EC2 files are disposable. Models, data, drift reports, canary results, and pipeline state live on S3.

Best interview framing: Start with the business problem and imbalance, then explain the metric choice, the bfloat16 failure/fix, the drift gate, the Day 30 retrain, the Day 39 rollback, and finally the live AWS recovery incidents. That sequence shows both ML depth and production ownership.

8. Visual Evidence

Existing project assets are embedded as evidence for PDF conversion. They keep the report visual without relying on external image generation.





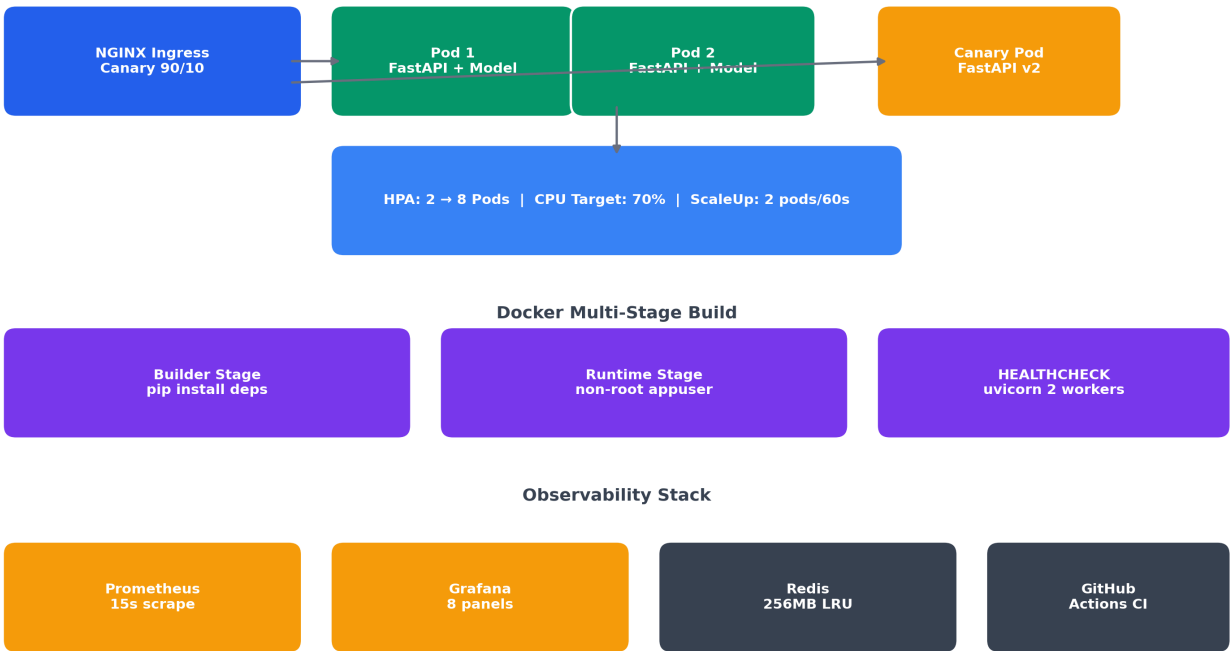
P053 — Production MLOps Architecture

Memory Yield Predictor — End-to-End Deployment



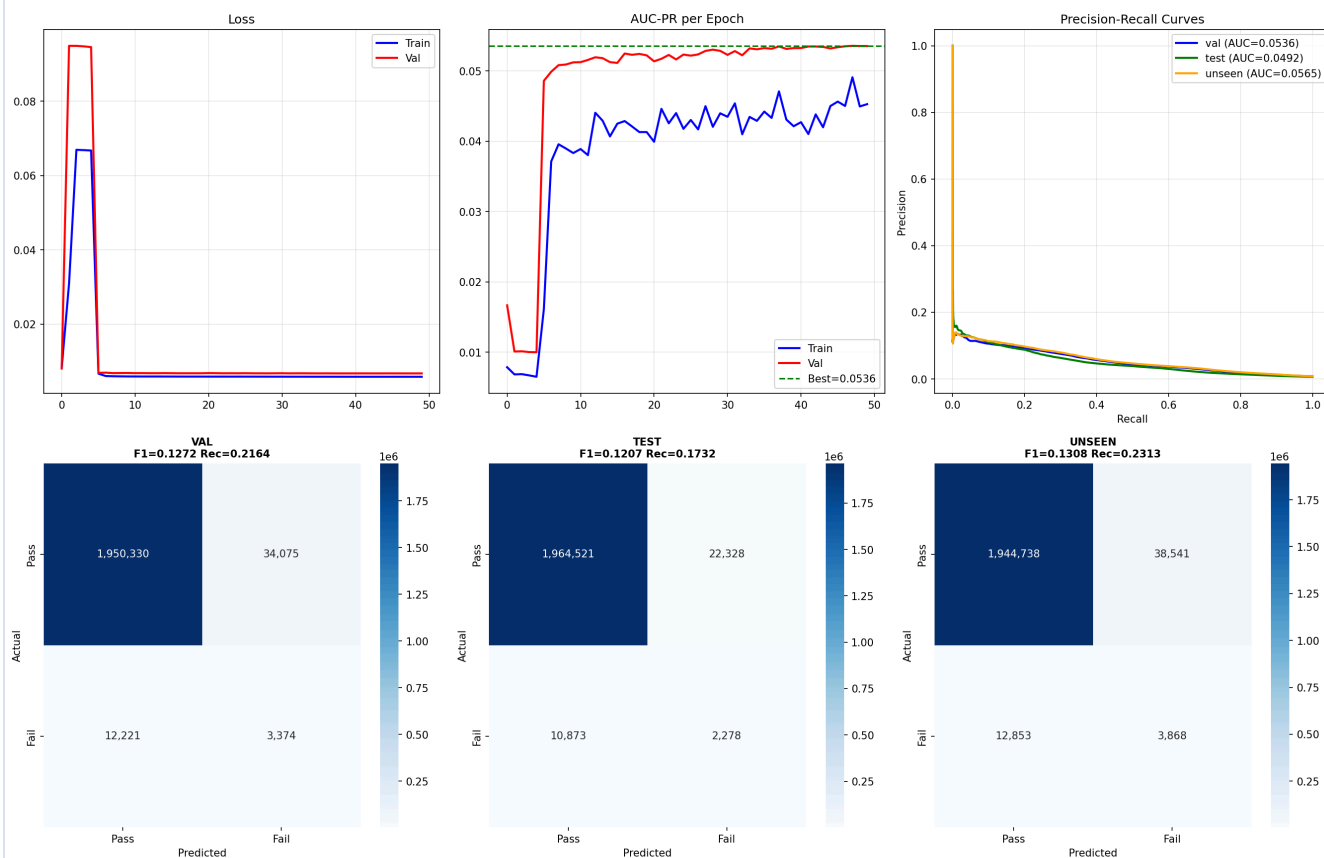
End-to-end architecture

Kubernetes Deployment Architecture



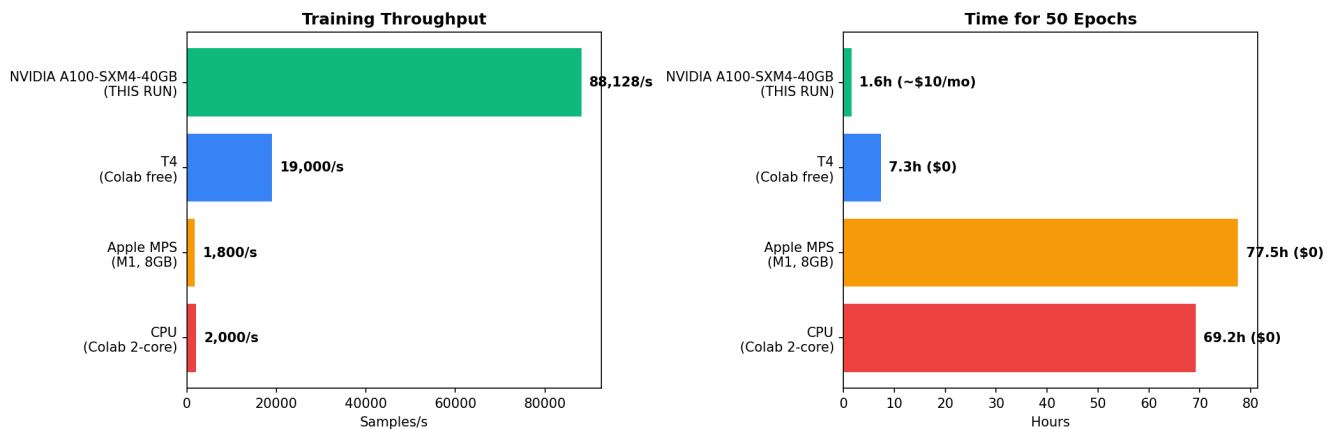
Deployment stack

Hybrid Transformer-CNN Results — NVIDIA A100-SXM4-40GB

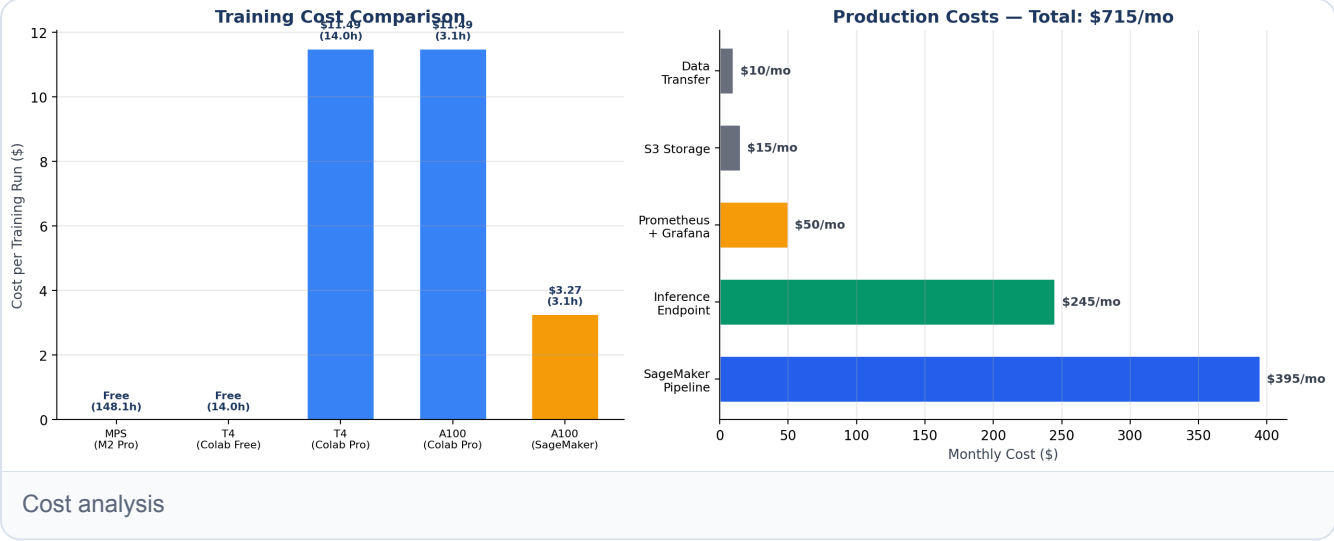


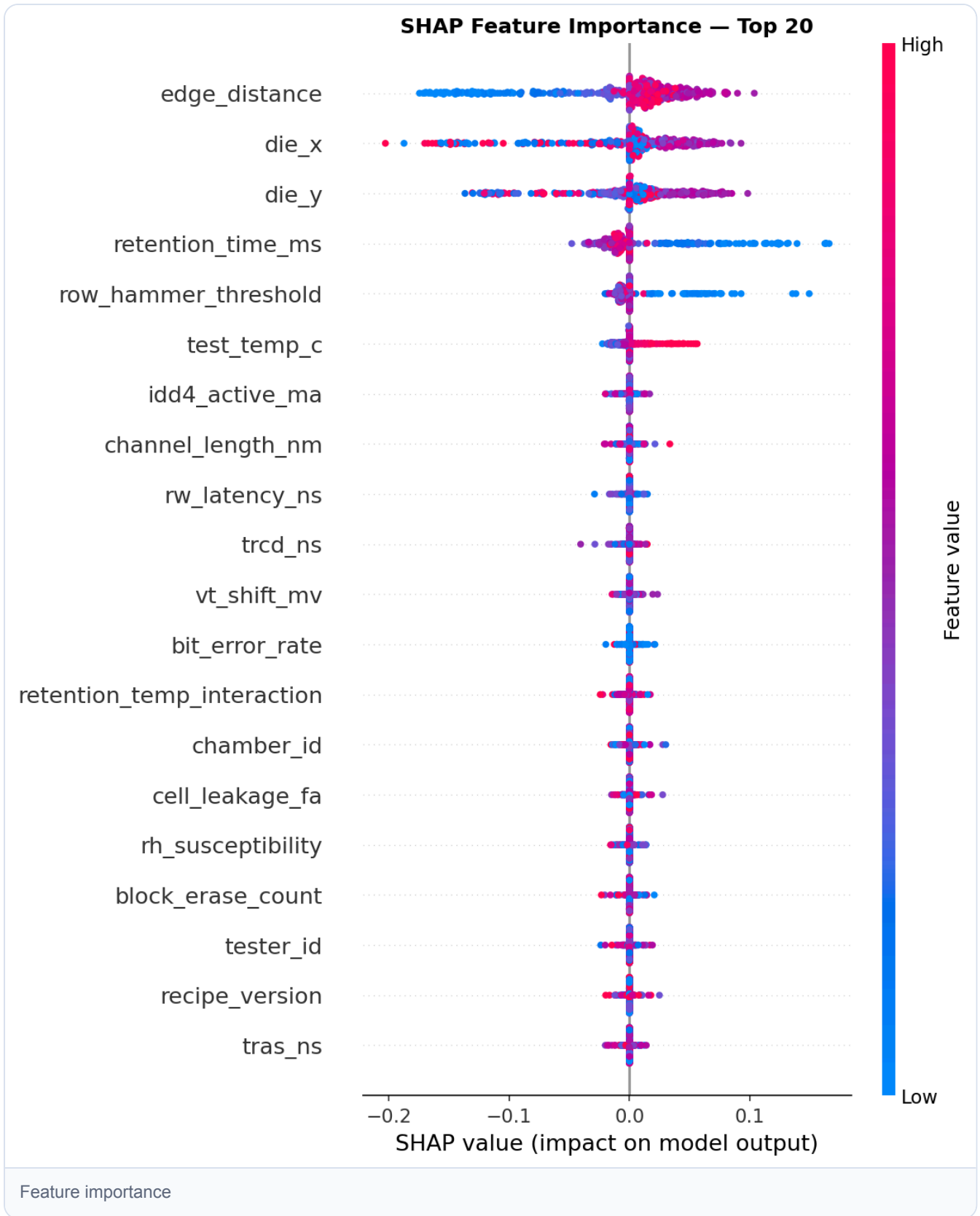
A100 training results

Hardware Benchmark — Why Cloud GPU is Non-Negotiable at Scale



Hardware benchmark





9. Update Plan for Future Days

1. Let daily pipeline finish new days and sync artifacts to S3.
2. Export or update local JSON artifacts: simulation timeline, live state, and metrics.
3. Run `python3 scripts/build_40day_learning_report.py`.

4. Open the HTML in a browser and print to PDF after the final days are complete.
5. Add final screenshots from GitHub Actions, S3, MLflow, Airflow, and Grafana if desired.

PDF tip: use browser print, A4, background graphics enabled, margins default or none. The stylesheet repeats table headers and avoids page breaks inside key cards.